PassFrame tuition

Statistics & Averages

Year 5 Accelerated — 11+ Lesson Worksheet (PassFrame Tuition)

Worked examples — tutor talks through these first

Easy: Find the mean of 12, 18, 15, 21, 9.
Method & answer: 12+18+15+21+9=75, 75÷5=15. Answer: 15.

Standard: A survey of 9 pupils' shoe sizes: 3, 4, 4, 5, 4, 6, 3, 5, 4. Find the mode and the median.
Method & answer: Mode = 4 (appears 4 times). Median: ordered 3,3,4,4,4,4,5,5,6 → the 5th value = 4.

Advanced: A pie chart shows how 150 pupils get to school: Walk=40%, Bus=30%, Car=20%, Cycle=10%. How many more pupils walk than cycle?
Method & answer: Walk = 40% of 150 = 60. Cycle = 10% of 150 = 15. Difference = 60 − 15 = 45.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Find the mean of 14, 22, 18, 26, 10.

2. Find the mean of 27, 33, 21.

3. Find the median of 16, 28, 9, 21, 17.

4. Find the median of 13, 29, 7, 35, 21.

5. Find the mode of 12, 17, 17, 23, 8, 17.

6. Find the mode of 21, 21, 28, 28, 28, 14.

7. Find the range of 24, 52, 36, 68, 40.

8. Find the range of 110, 80, 140, 70.

9. Which of these is the mode of 14, 14, 19, 26, 26, 26? A) 26 B) 14 C) 19 D) 20

10. A table shows pets owned across a school: Dogs=42, Cats=28, Fish=19, Birds=23. How many pets are there in total?

11. A café's coffee sales over 4 days: Mon=68, Tue=95, Wed=52, Thu=81. What is the mean number sold per day?

12. Select ALL that apply. A frequency table shows: 1 pet = 14 pupils, 2 pets = 19 pupils, 3 pets = 9 pupils. Which statements are true? A) 42 pupils were surveyed in total B) The mode number of pets owned is 2 C) The mode number of pets owned is 1 D) More pupils own 2 pets than own 3 pets

13. Five friends scored 68, 74, 81, 70, 77 in a test. What is their mean score? A) 74 B) 75 C) 73 D) 76

14. Find the mean of 16, 16, 16, 16, 26.

15. Find the range of 34, 67, 49, 18, 55.

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Find the median of 19, 32, 14, 25.

6. A bar chart shows sales at 4 shops: A=48, B=63, C=31, D=54. What is the mean sale per shop?

11. Which of these is the range of 87, 42, 69, 105, 58? A) 63 B) 105 C) 42 D) 61

2. Find the mean of 41, 53, 35, 63.

3. A library's book loans by class: Amy=23, Ben=31, Cara=18, Dan=36. What is the total number of loans?

4. Which of these is the median of 18, 37, 26, 55, 11? A) 26 B) 37 C) 18 D) 55

5. Find the range of 52, 104, 78, 33, 91.

7. Find the mode of 9, 15, 15, 19, 23, 15.

8. A pets survey shows: Dogs=37, Cats=29, Fish=21, Birds=13. If 12 cats are rehomed, how many pets are there altogether now?

9. Select ALL that apply. A class recorded favourite subjects: Maths=32, Art=21, Science=15. Which statements are true? A) 68 pupils were asked in total B) Maths is the mode (most popular) C) Science is the mode (most popular) D) More pupils chose Art than Maths

10. Find the mean of 24, 36, 28, 42.

12. Find the median of 21, 9, 31, 15, 41, 25.

13. A pie chart shows a survey of 84 pupils: half prefer football, a quarter prefer swimming. How many prefer something else?

14. Which of these is the mean of 38, 46, 54, 62? A) 50 B) 52 C) 48 D) 54

15. Find the range of 145, 198, 167, 178.

Standard / Hard — *work through together, answer 5 in the lesson* *15 min (1 min per question)*

1. Find the mean of 52, 44, 60, 36, 68.

2. Find the median of 38, 92, 24, 66, 74.

3. A pie chart shows 120 pupils' favourite sport: a third prefer football, a quarter prefer swimming, and the rest prefer something else. How many prefer something else?

4. A survey of 5 runners' times (minutes): 38, 45, 32, 41, 49. What is the range? A) 17 B) 11 C) 9 D) 49

5. Find the range of 136, 67, 124, 90, 82.

6. Six athletes scored 32, 38, 26, 44, 38, 50 points. Find the mode and the mean.

7. Select ALL that apply. A table shows a shop's weekly sales (£): Mon=88, Tue=118, Wed=68, Thu=138, Fri=128. Which statements are true? A) The mean daily sale is £108 B) The range of sales is £70 C) Thursday had the highest sales D) Wednesday had the lowest sales

8. Find the median of 46, 22, 58, 34, 70, 16.

9. A pie chart (360°) shows a school's transport choices: cycling gets 15% of the circle. What angle, in degrees, represents cycling?

10. Which of these is the mean of 48, 78, 58, 88? A) 68 B) 70 C) 66 D) 72

11. Find the mode of 96, 82, 96, 112, 96, 66.

12. A bakery sold 48 loaves in the morning and a quarter of that number in the afternoon. What was the mean number sold over the two sessions?

13. Find the range of 188, 98, 152, 124, 134.

14. Five athletes' long jump distances (cm) are 412, 458, 386, 501, 447. Find the median distance.

15. A pie chart shows that 65% of a class of 40 pupils prefer maths. How many pupils prefer maths? A) 26 B) 24 C) 28 D) 30